4.3 For each $n$, because $h$ is nonnegative, $n^{-1}\nu(B_n) \le \int_{B_n} h_n d\nu \le \int_{[0,1]} h\, d\nu < \infty$, where $B_n = \{x \in [0,1] \mid h(x) > n^{-1}\}$ and $\nu$ is the counting measure. Thus $B_n$ is a finite set for every $n \in \mathbb{N}$, and $B = \cup_{n\ge 1} B_n$ is countable.

4.4 (i) Since $\mu_1 \ll \mu_2 \ll \mu_3$, $\mu_3(A) = 0 \Rightarrow \mu_2(A) = 0 \Rightarrow \mu_1(A) = 0$, $\mu_1 \ll \mu_3$ follows. Given any two $\sigma$-finite measures $\mu$ and $\nu$ on some measurable space $(\Omega, \mathcal{F})$, with $\mu \ll \nu$, the following is true for any nonnegative measurable function $h$,

$$\int_\Omega h\, d\mu = \int_\Omega h \frac{d\mu}{d\nu}\, d\nu,$$

where $\frac{d\mu}{d\nu}$ is the Radon-Nikodym derivative, guaranteed by the Radon-Nikodym theorem, of $\mu$ w.r.t. $\nu$. This can be easily shown by first consider indicator functions, next check simple nonnegative measurable functions, then apply **MCT** to finish. For any $A \in \mathcal{F}$, let $h = I_A \frac{d\mu_1}{d\mu_2}$, and apply the above result to $\mu_2$, $\mu_3$, we get

$$\mu_1(A) = \int_\Omega h d\mu_2 = \int_\Omega h \frac{d\mu_2}{d\mu_3}\, d\mu_3 = \int_A \frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3}\, d\mu_3.$$

The existence and uniqueness (in the almost sure sense) of Radon-Nikodym derivatives are guaranteed, by applying Radon-Nikodym theorem, due to $\sigma$ finiteness of $\mu_1$, $\mu_2$, $\mu_3$. Therefore, $\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3}$, *a.e.*$(\mu_3)$.

(ii) We can verify that $\alpha\mu_1 + \beta\mu_2$ is a valid $\sigma$ -finite measure on $(\Omega, \mathcal{F})$, because $\alpha \ge 0, \beta \ge 0$. For any $A \in \mathcal{F}$,

$$\begin{aligned}(\alpha\mu_1 + \beta\mu_2)(A) &= \alpha\mu_1(A) + \beta\mu_2(A) = \alpha \int_A \frac{d\mu_1}{d\mu_3}\, d\mu_3 + \beta \int_A \frac{d\mu_2}{d\mu_3}\, d\mu_3 \\ &= \int_A \left(\alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3}\right) d\mu_3.\end{aligned}$$

Because $\alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3}$ is nonnegative measurable, $\alpha\mu_1 + \beta\mu_2 \ll \mu_3$ and its R-N derivative w.r.t. $\mu_3$ is $\alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3}$ .

(iii) For any $A \in \mathcal{F}$, let $h = I_A \cdot \left(\frac{d\mu}{d\nu}\right)^{-1}$, who is valid because $\frac{d\mu}{d\nu} > 0$, *a.e.*$(\nu)$. Apply the result in (i)

$$\begin{aligned}\nu(A) &= \int_\Omega I_A d\nu = \int_\Omega I_A \left(\frac{d\mu}{d\nu}\right)^{-1} \frac{d\mu}{d\nu}\, d\nu \\ &= \int_\Omega h\frac{d\mu}{d\nu}\, d\nu = \int_\Omega h d\mu \\ &= \int_A \left(\frac{d\mu}{d\nu}\right)^{-1} d\mu.\end{aligned}$$

Thus $\nu \ll \mu$ and $\dfrac{d\nu}{d\mu} = \left(\dfrac{d\mu}{d\nu}\right)^{-1}$, $a.e.(\mu)$.

(iv) Countable additivity of $\mu$ can be verified using the result in problem 2.20, $\mu$ is then a valid measure in $(\Omega, \mathcal{F})$. Since the sum of a nonnegative sequence is 0 iff every term in the sum is 0, it's easy to see $\mu \ll \nu \Leftrightarrow \mu_n \ll \nu$ for all $n \in \mathbb{N}$. In this case, for any $A \in \mathcal{F}$, by Corollary 2.3.5,

$$\mu(A) = \sum_{i=1}^{\infty} \int_A \alpha_n \frac{d\mu_n}{d\nu}\, d\nu = \int_A \left( \sum_{i=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu} \right) d\nu,$$

where $\sum_{i=1}^{\infty} \alpha_n \dfrac{d\mu_n}{d\nu} \geq 0$, $a.e.(\nu)$. So $\dfrac{d\mu}{d\nu} = \sum_{i=1}^{\infty} \alpha_n \dfrac{d\mu_n}{d\nu}$, $a.e.(\nu)$.
As we said earlier, $\mu(B) = 0 \Rightarrow \mu_n(B) = 0$ for all $n \geq 1$ since $\alpha_i$'s are all positive real numbers. Furthermore, $\mu_n(B_n) = 0$ and $D = \cap B_n \Rightarrow \mu_n(D) = 0$ and then $\mu(D) = 0, \nu(D^c) = \nu(\cup B_n^c) \leq \sum \nu(B_n^c) = 0$. Thus $\mu \perp \nu$ iff $\mu_n \perp \mu$ for all $n \geq 1$.

4.5 Let Lebesgue measure restricted to $\mathbb{R}^+$, denoted by $m_1$, i.e. $m_1(A) = m_1(A \cap \mathbb{R}^+)$, $\forall A \in \mathcal{B}(\mathbb{R})$. With similar manner, define the counting measure $\lambda$ restricted on $\mathbb{N}$ as $\lambda_1$.

(a) Define $\mu_a(A) = \mu(A \cap \mathbb{R}^+)$, $\forall A \in \mathcal{B}(\mathbb{R})$, then $\mu_a \ll m_1$. $\mu_a$ is the absolute continuous part of $\mu$ w.r.t. $\mu$, because we can find $B = \mathbb{R}^+$ such that $(\mu - \mu_a)(B) = 0$ and $\nu(B^c) = 0$. On the other hand, $\nu \ll m_1$ and $\dfrac{d\nu}{dm_1} = e^{-x} > 0$, $a.e.(m_1)$. By proposition 4.1.2, $\mu_a \ll m_1 \ll \nu$ and $\dfrac{d\mu_a}{d\nu} = \dfrac{d\mu_a}{dm_1}\left(\dfrac{d\nu}{dm_1}\right)^{-1} = \dfrac{1}{\sqrt{2\pi}} e^{-x^2/2+x}$, $a.e.(\nu)$.

(b) From the proof in (a), by Proposition 4.1.2, we know that $\mu \ll m_1 \ll m \ll \nu$, because $\dfrac{d\nu}{dm} > 0$, $a.e.(m)$ and $m_1 \ll m$ is obvious. We know that Lebesgue decomposition is unique, so $\mu$ is the absolute continuous part of $\mu$ w.r.t. $\nu$. and $\dfrac{d\mu}{d\nu} = \dfrac{d\mu}{dm}\left(\dfrac{d\nu}{dm}\right)^{-1} = \sqrt{2\pi} e^{-x+x^2/2} I\{x > 0\}$, $a.e.(\nu)$.

(c) Clearly, $\mu_1 \ll m \ll \nu$ because the cauchy density is strictly positive on $\mathbb{R}$. $\nu(\{0\} \cup \mathbb{N}) = 0, \mu_2\{(\{0\} \cup \mathbb{N})^c\} = 0$ says that $\mu_2 \perp \nu$. So $\mu_1$ is the absolute continuous part of $\mu_1 + \mu_2$ w.r.t. $\nu$ and $\dfrac{d\mu_1}{d\nu} = \dfrac{d\mu_1}{dm}\left(\dfrac{d\nu}{dm}\right)^{-1} = \sqrt{\dfrac{\pi}{2}}(1+x^2)e^{-x^2/2}$, $a.e.(\nu)$.

(d) Consider $\mu_{2s}(A) = \mu_2(A \cap \{0\})$, $\forall A \in \mathcal{B}(\mathbb{R})$, who automatically defines another measure $\mu_{2a}(A) = \mu_2(A \cap \mathbb{N})$, $\forall A \in \mathcal{B}(\mathbb{R})$. Since $(\mu_1 + \mu_{2s})(\mathbb{N}) = 0, \nu(\mathbb{N}^c) = 0$ and $\nu(A) = 0 \Rightarrow A \cap \mathbb{N} = \emptyset \Rightarrow \mu_{2a}(A) = 0$, $\mu_{2a} \ll \nu$ and $(\mu_1 + \mu_{2s}) \perp \nu$. So $\mu_{2a}$ is the absolute continuous part of $(\mu_1 + \mu_2)$ w.r.t. $\nu$. and $\mu_{2a} \ll \lambda_1 \ll \nu$ where $\lambda_1$ is counting measure.

$$\begin{aligned}
\frac{d\mu_{2a}}{d\nu}(\omega) &= \frac{d\mu_{2a}}{d\lambda_1}\frac{d\lambda_1}{d\nu} \\
&= \left( \sum_{\omega \in \mathbb{N}} I\{\omega = k\}(1-p)^{k-1}p \right)^{-1} \cdot \left( \sum_{\omega \in \mathbb{N}} I\{\omega = k\} \frac{e^{-1}}{k!} \right) \\
&= \sum_{\omega \in \mathbb{N}} \frac{(1-p)^{1-k}}{epk!} I\{\omega = k\},\ a.e.(\nu).
\end{aligned}$$

(e) Define $\mu_{2s}(A) = \mu_2(A \cap \{0\}), \mu_{2a}(A) = \mu_2(A \cap \mathbb{N}), \forall A \in \mathcal{B}(\mathbb{R})$. Since $\mu_{2s}(\mathbb{R} \setminus \{0\}) = 0, (\nu_1 + \nu_2)(\{0\}) = 0$, $\mu_{2s} \perp \nu_1 + \nu_2$. Then again, $\mu_1 \ll \nu_1, \mu_{2a} \ll \nu_2 \Rightarrow (\mu_1 + \mu_{2a}) \ll (\nu_1 + \nu_2)$. Thus $\mu_1 + \mu_{2a}$ is absolute continuous part of $\mu_1 + \mu_2$ w.r.t. $\nu_1 + \nu_2$.

Now because $\mu_1 \perp \nu_2, \mu_{2a} \perp \nu_1, \mu_1 \ll \nu_1, \mu_{2a} \ll \nu_2$, for any $A \in \mathbb{N}$,

$$
\begin{aligned}
&(\mu_1 + \mu_{2a})(A) = \mu_1(A) + \mu_{2a}(A) \\
&= \int_A \frac{d\mu_1}{d\nu_1}\, d\nu_1 + \int_A \frac{d\mu_{2a}}{d\nu_2}\, d\nu_2 \\
&= \int_A \frac{d\mu_1}{d\nu_1} I_{\mathbb{N}^c} d\nu_1 + 0 + \int_A \frac{d\mu_{2a}}{d\nu_2} I_{\mathbb{N}}\, d\nu_2 + 0 \\
&= \int_A \frac{d\mu_1}{d\nu_1} I_{\mathbb{N}^c} d\nu_1 + \int_A \frac{d\mu_1}{d\nu_1} I_{\mathbb{N}^c} d\nu_2 + \int_A \frac{d\mu_{2a}}{d\nu_2} I_{\mathbb{N}}\, d\nu_2 + \int_A \frac{d\mu_{2a}}{d\nu_2} I_{\mathbb{N}}\, d\nu_1 \\
&= \int_A \frac{d\mu_1}{d\nu_1} I_{\mathbb{N}^c} d(\nu_1 + \nu_2) + \int_A \frac{d\mu_{2a}}{d\nu_2} I_{\mathbb{N}}\, d(\nu_1 + \nu_2) \\
&\therefore \frac{d(\mu_1 + \mu_{2a})}{d(\nu_1 + \nu_2)} = \frac{d\mu_1}{d\nu_1}\, I_{\mathbb{N}^c} + \frac{d\mu_{2a}}{d\nu_2}\, I_{\mathbb{N}} \\
&= \sqrt{\frac{\pi}{2}}\, (1 + x^2) e^{-x^2/2} I\{x \in \mathbb{N}^c\} + \sum_{x \in \mathbb{N}} \frac{1}{ep(1-p)^{k-1} k!} I\{x = k\},\ a.e.(\nu_1 + \nu_2).
\end{aligned}
$$

(f) Clearly, $\mu \ll \nu$, and for any $A \in \mathcal{B}(\mathbb{R})$,

$$
\mu(A) = \sum_{\omega \in A \cap \{0, \dots, 10\}} \binom{10}{\omega} 2^{-10} = \int_A \binom{10}{\omega} \frac{e\,\omega!}{2^{10}} I_{\{0,\dots,10\}}(\omega) d\nu(\omega),
$$

so $\dfrac{d\mu}{d\nu}(\omega) = \dbinom{10}{\omega} \dfrac{e\,\omega!}{2^{10}} I_{\{0,\dots,10\}}(\omega),\ a.e.(\nu)$.